A LIGHTWEIGHT DEEP LEARNING FRAMEWORK FOR CALCIFICATION'S RISK PREDICTION IN MAMMOGRAPHY
(Một khung học sâu nhẹ dự đoán nguy cơ từ tổn thương vôi hóa trên ảnh nhũ ảnh)

Tran Mai Phuong[1,3,4]_2352960_Computer Science, Dong Minh Nghia[2,3,4]_2252525_Biomedical Engineering, Do Thai Binh Ca[2,3,4]_2252086_Biomedical Engineering, Nguyen Thi Thanh Truc[2,3,4]_2152330_Biomedical Engineering, Le Nhat Tan[2,4,*]
[1] Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City, Vietnam
[2] Faculty of Applied Science, Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City, Vietnam
[3] Office for International Study Programs, Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City, Vietnam
[4] Vietnam National University Ho Chi Minh City, Linh Trung Ward, Thu Duc District, Ho Chi Minh City, Vietnam
* Corresponding author: lenhattan@hcmut.edu.vn

Abstract

Breast cancer was the most common cancer in women in 157 countries out of 185 in 2022. Early detection through mammography plays a crucial role in improving survival rates. However, risk prediction remains a significant challenge. Unnecessary callbacks and missed cancer cases contribute to both increased psychological distress and economic burden in healthcare systems. In this paper, we propose a lightweight and interpretable pipeline for calcification's risk classification using frozen features from MONAI's DenseNet121 and an XGBoost classifier. Our approach avoids costly retraining while still maintaining strong performance. To enhance clinical relevance, we further apply Grad-CAM to visualize model attention over mammogram regions, increasing transparency and trust. Our method achieves 93.6% sensitivity, 86.1% precision, and 89.7% F1-score on the CBIS-DDSM dataset, with an AUROC of 86.7%. These results are competitive with benchmarks, while offering significant improvements in computational efficiency and interpretability, making the proposed system suitable for deployment in real-time or resource-constrained environments.

Keywords: *Breast cancer, Mammography, Calcification's risk prediction, Deep learning, DenseNet121, XGBoost, Grad-CAM, Transfer learning, breast image analysis*

1. Introduction

Breast cancer remains the most commonly diagnosed cancer among women worldwide and is a leading cause of cancer-related mortality. According to the World Health Organization, in 2022 alone, an estimated 2.3 million women were diagnosed with breast cancer, resulting in approximately 670,000 deaths globally [1]. This global burden underscores the urgent need for effective screening and diagnostic strategies to enable early detection and timely intervention.

Mammography is widely used for screening and has been shown to reduce breast cancer mortality. However, its application is often accompanied by a significant clinical challenge: the risk prediction, a situation in which patients are asked to return for additional imaging following a suspicious finding [2]. While callbacks can improve diagnostic certainty and prevent missed cancers, they are also associated with high false-positive rates. These false positives contribute to unnecessary anxiety, increased healthcare costs, and workload burdens on clinical staff [3]. Conversely, false negatives where malignancies are missed pose a life-threatening risk due to delayed treatment [4]. Therefore, any improvement in the diagnostic pipeline must strike a careful balance between sensitivity and specificity.

Recently, artificial intelligence (AI), particularly deep learning (DL), has shown potential to support radiologists in improving diagnostic accuracy. Multiple studies have demon- strated the effectiveness of convolutional neural networks (CNNs) and transformer-based architectures in mammographic image classification [5][6]. Some researches displayed high results in this field. For instance, IMPA-ResNet50 [7] achieves 98.56% sensitivity, and 98.68% specificity. Another research, Jafari et al. [8], used pre-trained CNNs such as ResNet50 and EfficientNet for feature extraction got 94.70% and 97% in recall and precision respectively.

Despite these advancements, existing DL pipelines often face two critical limitations when deployed in real-world screening workflows: high computational cost, due to end-to-end training and large model architectures, and limited interpretability, as many deep networks act as black boxes with unclear decision boundaries. These constraints hinder their integration into point-of-care systems, especially in resource-constrained environments.

In this study, we propose a lightweight and interpretable pipeline for calcification's risk classification in mammography. We utilize frozen feature representations from MONAI's DenseNet121. These frozen features are classified using XGBoost. Additionally, we applied Grad-CAM to provide pixel-level interpretability over the original images. Our results demonstrate that this hybrid approach achieves competitive performance while significantly reducing computational cost and improving explainability, making it a strong candidate for real-world risk assessment and triage applications.

2. Methods and Implementation

The general pipeline of our proposed method is displayed in Figure 1.

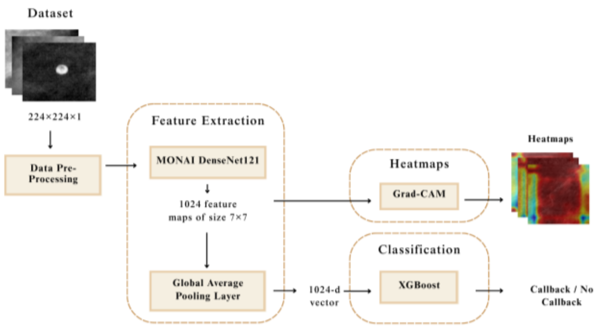


Figure 1: Overview of the Proposed Pipeline

The pipeline consists of preprocessing cropped mammogram inputs, feature extraction using a frozen MONAI DenseNet121, dimensionality reduction via global average pooling, and binary classification through XGBoost. Grad-CAM is applied to the final convolutional feature maps to produce visual explanations for model decisions.

a. Dataset & data pre-processing

In this study, we utilize the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS- DDSM), a publicly available and widely adopted benchmark dataset for mammographic image analysis curated by the Cancer Imaging Archive (TCIA) [9]. This data set is a clean, standardized and well-annotated version of the original DDSM dataset, consisting of digitized film mammograms with pixel-level lesion annotations and pathology-confirmed diagnostic labels.

In this study, we only used cropped images for training and testing. Therefore, patients with only full images were eliminated from the final dataset. After filtering, it resulted in 1870 cropped grayscale images. Each image was resized to 224×224 pixels and normalized to match the input requirements of the MONAI DenseNet121 model used for feature extraction.

Table 1: Summary of Label Types and Class Distribution in the Final Dataset

Label Type	Class Description	No. of Images
Pathology	Benign without callback	540
	Benign	658
	Malignant	672
Binary	No callback	540
	Callback	1330
Total		1870

The pathology field in the dataset contains diagnostic labels that describe the nature of the abnormality. As shown in Table 1, the original annotations include three categories: benign without callback, benign, and malignant. For the purpose of binary classification, we grouped the latter two categories into a single callback class, while maintaining the benign without callback cases as the no callback class. This results in a class distribution of 540 no-callback cases and 1330 callback cases. This binary label serves as the ground-truth target for the supervised classification task.

We can observe an imbalance in the labels, thus, we already applied class weight in order to handle this problem. Since the original cropped images are of high quality and well-centered on the lesion areas, no additional filtering or flipping was applied during preprocessing.

b. Feature Extraction Model

To extract meaningful representations from mammogram images in a computationally efficient manner, we adopt DenseNet121 as implemented in the Medical Open Network for AI (MONAI) framework [10] [11]. MONAI is a PyTorch-based platform tailored for deep learning in healthcare imaging, offering optimized pretrained models for various medical tasks.

DenseNet121 belongs to the family of Densely Connected Convolutional Networks, which introduces direct connections between all layers within the same dense block [10]. Unlike traditional CNNs where each layer feeds only into the next, DenseNet connects each layer to every other layer in a feed-forward manner. Formally, the l^{th} layer receives the feature maps of all preceding layers as input:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}]) \quad (1)$$

where $H_l(\cdot)$ denotes a composite function of batch normalization, ReLU activation, and 3×3 convolution; $[x_0, x_1, \ldots, x_{l-1}]$ represents the concatenation of feature maps from layers 0 to $l - 1$.

In MONAI, DenseNet121 is pretrained on large-scale medical image datasets, capturing domain-specific patterns that are relevant for diagnostic imaging tasks. In our proposed approach, we utilize DenseNet121 purely as a feature extractor. All weights in the network are frozen to reduce training time and avoid overfitting, particularly given the limited size of annotated medical imaging datasets. Each input mammogram is converted to grayscale, resized to 224×224 pixels, and normalized. The image is then passed through the DenseNet backbone, producing a feature map of shape (C, H, W), where $C = 1024$ represents the number of channels. To convert these high-dimensional feature maps into fixed-length vectors suitable for classical machine learning models, we apply global average pooling:

$$f_i = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} x_{i,h,w}, i = 1, \ldots, C \quad (2)$$

This operation yields a 1024-dimensional vector that summarizes the spatial information from each feature channel. The resulting feature vectors are stored and used as input for the classification stage using XGBoost.

c. Classification Model

After extracting 1024-dimensional feature vectors from each mammogram, we perform classification using Extreme Gradient Boosting (XGBoost) [12]. It builds an additive model in a forward stage-wise manner by optimizing a regularized objective function. At iteration t, the model predicts:

$$\hat{y}_i^{(t)} = \sum_{k=1}^{t} f_k(x_i), \qquad f_k \in \mathcal{F} \quad (3)$$

where $\mathcal{F}$ is the space of regression trees and each f_k corresponds to a decision tree added at iteration k. The objective function combines a convex loss function $\mathcal{L}$ and a regularization term Ω to prevent overfitting:

$$\mathcal{L}_{total} = \sum_{i=1}^{n} \mathcal{L}(y_i, \hat{y}_i) + \sum_{k=1}^{t} \Omega(f_k) \quad (4)$$

XGBoost is particularly well-suited for our problem for three main reasons. First, it is computationally efficient and trains quickly on small- to medium-sized datasets. Second, it handles imbalanced classes well through parameterized loss weighting. Third, its tree-based structure offers transparency, allowing feature importance analysis and integration with interpretability tools. We train the XGBoost classifier binary labels corresponding to callback and non-callback cases.

d. Evaluation Metrics

The result in both cross-validation methods are performed in five metrics: Sensitivity (recall) Precision, F1-Score, Accuracy and AUROC. While accuracy reflects overall correctness, it may be misleading in imbalanced datasets like in our case.

In medical contexts, Recall is critical as it measures the ability to correctly identify patients who truly have high risk of calcification. Precision helps reduce unnecessary callbacks, minimizing patient stress and healthcare burden. F1-Score balances both, while AUROC assesses the model's discrimination capability across thresholds. Together, these metrics offer a more meaningful evaluation than accuracy alone for high-stakes medical decision-making.

3. Results and Discussion

a. Benchmarking

To evaluate the effectiveness of our proposed method, we compare it against several recent deep learning-based approaches for mammogram classification, focusing on sensitivity (recall), precision, F1-score, accuracy, and AUROC. Table 2

Table 2: Comparison of Classification Performance Across Models and Mammography Datasets

References	Dataset(s)	Sensitivity	Precision	F1-Score	Accuracy	AUC-ROC
CLAM [13]	Chinese Mammography Database	78.4 ± 1.9	78.8 ± 2.0	78.6 ± 2.0	78.4 ± 1.9	0.848 ± 0.015
IMPA-ResNet50 [7]	CBIS-DDSM	96.61	98.68	97.65	98.32	-
IMPA-ResNet50 [7]	MIAS	97.61	98.30	97.10	98.88	99.24
DenseNet201-C [14]	MINI-MIAS	94.58	-		92.73	-
Ours	CBIS-DDSM	93.6	86.1	89.7	84.7	86.7

summarizes this comparison using results reported on public datasets, including CBIS-DDSM, MIAS, MINI-MIAS, and the Chinese Mammography Database.

CLAM [13], a weakly supervised multiple instance learning framework based on ResNet and EfficientNet, achieved a sensitivity of 78.4% and AUROC of 0.848 on classification of benign and malignant on the Chinese Mammography Database. Although the model benefits from attention-based interpretability and multi-view analysis, it underperforms in core diagnostic metrics and requires costly full-model training.

IMPA-ResNet50 [7] applies hybrid fine-tuning and dropout strategies to boost classification performance. It demonstrates strong performance on CBIS-DDSM and MIAS in terms of classifying benign and malignant, with sensitivity above 96% and AUROC as high as 0.9924, but the model is computationally intensive and lacks lightweight deployment feasibility.

The DenseNet201-C model [14] adopts a fine-tuned DenseNet architecture and achieves high accuracy (92.73%) on the MINI-MIAS dataset in the task of classifying normality and abnormality, with sensitivity of 94.58% and specificity of 91.67%. While it demonstrates excellent diagnostic capability, its training requires significant time and GPU resources.

Our pipeline achieves a competitive sensitivity of 93.6%, a precision of 86.1%, and an F1 score of 89.7% on CBIS-DDSM. The AUROC of 0.867 further confirms the discriminative capability of the model.

Regarding to the computational cost, our proposed pipeline offers substantial improvements in computational efficiency. DenseNet201-C, which applies full fine-tuning on a deep architecture, requires an average of 1,620 seconds for training on a high-end GPU (Tesla P100) and exhibits inference times often exceeding 1 second per image [14]. Similarly, IMPA-ResNet50 achieves impressive accuracy and AUROC but demands end-to-end training and fine-tuning strategies, making it one of its main limitations [7]. Our method leverages frozen features from MONAI's DenseNet121, eliminating the need for backpropagation through the feature extractor. Classification is handled by a lightweight XGBoost model, reducing the total training time to under 2 minutes on a CPU, and enabling inference times below 150 milliseconds per image.

Our framework strikes a better trade-off between performance, transparency, and computational cost, making it ideal for real-time screening and deployment in resource-constrained environments.

b. Qualitative Analysis

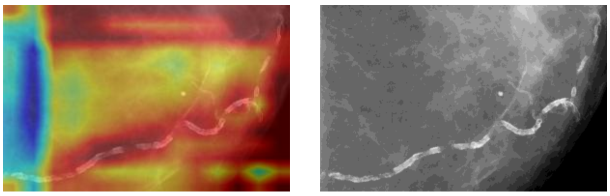
Breast cancer screening often relies on the early detection of microcalcifications that are related to the presence of malignant lesions [15]. Radiologists frequently use these subtle visual cues to assess risk, particularly in cases where mass formation is not yet apparent [15]. In this study, it is critical not only to achieve high predictive accuracy, but also to provide visual explanations that align with radiological understanding.

We used Gradient-weighted Class Activation Mapping (Grad-CAM) [16] to visualize the spatial focus of the models. Grad-CAM produces heatmaps that highlight the regions in an input image that contribute most strongly to the model's classification decision. By overlaying these heatmaps on the original mammogram images, we can assess whether the model is paying attention to diagnostically meaningful regions, such as clustered calcifications, rather than irrelevant artifacts.

For qualitative evaluation, we selected three representative cases from the CBIS-DDSM dataset:

- Case 1: Vascular - Benign not Callback
- Case 2: Pleomorphic - Benign with Callback
- Case 3: Fine Linear Branching – Malignant

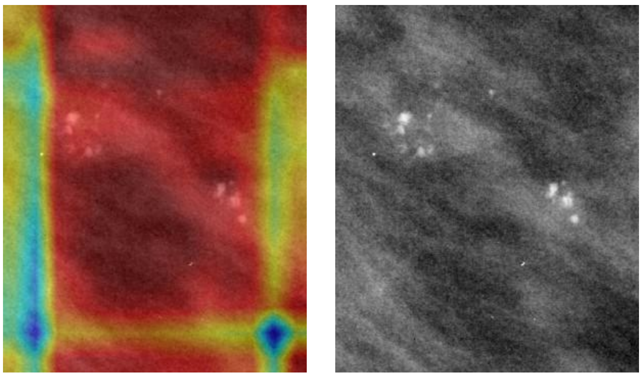
1) Case 1 - Vascular: Vascular calcifications are typically seen as dense, linear, parallel, circuitous or tram-track like calcifications. They are usually not oriented in the direction of the duct towards the nipple-areolar complex. [17]



(a) Heatmap (b) Original mammogram

Figure 2: Case 1 - Vascular Calcification

As shown in Figure 2, the Grad-CAM heatmap (2a) produced by our model focuses precisely on the serpentine structure of the vascular calcification, which is also visible in the original mammogram (2b). The



(a) Heatmap (b) Original mammogram

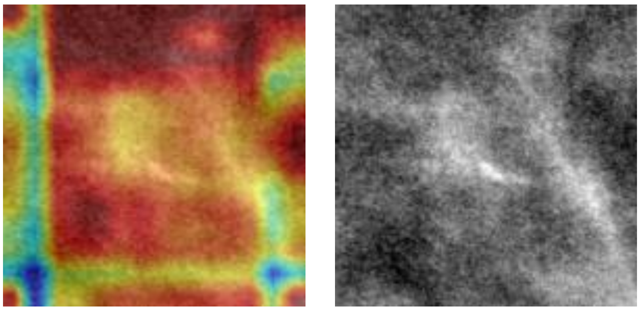
Figure 3: Case 2 - Pleomorphic Calcification

activation is concentrated along the curved, high-contrast vessel-like formation, while surrounding tissues and background noise are largely ignored. This attention pattern confirms that the model is learning to distinguish benign vascular features from suspicious ductal or pleomorphic patterns.

2) Case 2 - Pleomorphic: Vary in size and shape, and are clustered in a specific area of the breast, with variable shapes (”shards of glass” or ”crushed stone”), generally <0.5 mm. [17]

In Figure 3, the original mammogram (3b) clearly shows clustered, irregular calcifications in the center of the region of interest. The Grad-CAM heatmap (3a) highlights this area with strong activation. This suggests that the model has correctly identified the location and morphology of the pleomorphic cluster as the most informative for its malignant classification.

3) Case 3 - Fine Linear Branching: Fine linear branch types are described as thin (<0.5 mm), linear or branching calcifications (casting) within the conduits, highly suggestive of high-grade ductal carcinoma in situ (DCIS). [17]



(a) Heatmap (b) Original mammogram

Figure 4: Case 3 - Fine linear branching calcification

In this case (Figure 4), the original mammogram (4b) withnclearly visible fine linear branching calcifications at the center. The Grad-CAM heatmap (4a) generated by our proposed pipeline. Notably, the activation is highly concentrated around the central ductal structures, aligning well with the region of pathological interest. The model successfully focuses on the calcification clusters, avoiding irrelevant background noise.

c. Discussion

Although our model's performance is slightly lower than that of fully fine-tuned diagnostic models, it specifically addresses calcification risk classification. Unlike diagnosis-focused approaches that aim to determine malignancy, our model predicts whether further investigation is warranted, regardless of pathology. This is particularly important in breast screening workflows where even benign findings may necessitate recall due to uncertainty or atypical presentations. As such, our framework provides actionable triage support, aligning more closely with operational demands in real-world clinical practice.

Despite the promising results, several challenges remain. First, the CBIS-DDSM dataset has an inherent imbalance between callback and non-callback classes, which risks biasing the classifier. While this was partially mitigated through class weighting and threshold tuning, it may still affect specificity and generalization.

Second, the current setup treats calcification's risk as a binary classification task. However, in real-world scenarios, it would be more informative to distinguish between “benign no call-back”, “benign with callback”, and “malignant”, suggesting a need for multi-class or hierarchical classification models.

Finally, while our model is efficient and interpretable, its AUROC is slightly lower than some fully fine-tuned benchmarks. Further optimization, possibly through semi-supervised learning or multi-branch fusion, may be needed to close this gap. Feature selection should also be applied to reduce redundancy in the extracted embeddings and enhance the discriminative capacity of the classifier.

In general, our findings support the feasibility of a resource-efficient and explainable pipeline for mammogram triage, paving the way for wider adoption in routine screening workflows.

4. Conclusions

The study presents a lightweight and interpretable pipeline for calcification's risk classification in mammography, combining frozen MONAI DenseNet121 features with a calibrated XGBoost classifier. The approach achieves competitive sensitivity and precision while dramatically reducing computational cost.Grad-CAM visualizations further enhance model transparency by localizing diagnostically relevant regions.

We demonstrated that our method delivers strong sensitivity and precision, aligning with the clinical priority of minimizing missed cancers while reducing unnecessary callbacks. Importantly, the proposed pipeline avoids the computational overhead of end-to-end deep learning models, making it highly suitable for deployment in resource-constrained environments such as low-power diagnostic devices or real-time triage systems. Our results demonstrate the method's suitability for real-time or resource-limited deployments, offering an effective alternative to fully trainable deep learning models. Nonetheless, challenges remain, including class imbalance, binary task framing, and slightly lower AUROC compared to heavily fine-tuned baselines.

Future work might address these limitations through multi-class extension, advanced augmentation, and hybrid architectures. Overall, our pipeline offers a promising foundation for scalable and explainable AI solutions in breast cancer screening.

References

[1] World Health Organization, "Breast cancer." https://www.who.int/news-room/fact-sheets/detail/breast-cancer, 2022. Accessed: 2025-05-24.

[2] W. P. Evans, "Breast cancer screening: successes and challenges," 2012.

[3] R. A. Hubbard, K. Kerlikowske, C. I. Flowers, B. C. Yankaskas, W. Zhu, and D. L. Miglioretti, "Cumulative probability of false-positive recall or biopsy recommendation after 10 years of screening mammography: a cohort study," Annals of Internal Medicine, vol. 155, pp. 481–492, Oct 2011. Erratum in: Ann Intern Med. 2014 May 6;160(9):658.

[4] C. D. Lehman, R. F. Arao, B. L. Sprague, J. G. Lee, D. S. Buist, K. Kerlikowske, L. M. Henderson, T. Onega, A. N. Tosteson, G. H. Rauscher, and D. L. Miglioretti, "National performance benchmarks for modern screening digital mammography: Update from the breast cancer surveillance consortium," Radiology, vol. 283, pp. 49–58, Apr 2017. Epub 2016 Dec 5.

[5] A. Rodriguez-Ruiz, K. L˚ang, A. Gubern-Merida, M. Broeders, G. Gennaro, P. Clauser, T. H. Helbich, M. Chevalier, T. Tan, T. Mertelmeier, M. G. Wallis, I. Andersson, S. Zackrisson, R. M. Mann, and I. Sechopoulos, "Stand-alone artificial intelligence for breast cancer detection in mammography: Comparison with 101 radiologists," JNCI: Journal of the National Cancer Institute, vol. 111, pp. 916–922, 03 2019.

[6] H.-E. Kim, H. H. Kim, B.-K. Han, K. H. Kim, K. Han, H. Nam, E. H. Lee, and E.-K. Kim, "Changes in cancer detection and false-positive recall in mammography using artificial intelligence: a retrospective, multireader study," The Lancet Digital Health, vol. 2, no. 3, pp. e138–e148, 2020.

[7] E. H. Houssein, M. M. Emam, and A. A. Ali, "An optimized deep learning architecture for breast cancer diagnosis based on improved marine predators algorithm," Neural computing and applications, vol. 34, no. 20, pp. 18015–18033, 2022.

[8] Z. Jafari and E. Karami, "Breast cancer detection in mammography images: A cnn-based approach with feature selection," Information, vol. 14, no. 7, p. 410, 2023.

[9] Awsaf49, "Cbis-ddsm breast cancer image dataset." https://www.kaggle.com/datasets/awsaf49/cbis-ddsm-breast-cancer-image-dataset/data, 2021. Accessed: 2025-05-24.

[10] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4700–4708, 2017.

[11] M. J. Cardoso, W. Li, T. Brown, et al., "MONAI: An open-source framework for deep learning in healthcare," arXiv preprint arXiv:2211.02701, 2022.

[12] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794, ACM, 2016.

[13] M. Y. Lu, D. F. Williamson, T. Y. Chen, R. J. Chen, M. Barbieri, and F. Mahmood, "Data-efficient and weakly supervised computational pathology on whole-slide images," Nature biomedical engineering, vol. 5, no. 6, pp. 555–570, 2021.

[14] X. Yu, N. Zeng, S. Liu, and Y. Zhang, "Utilization of densenet201 for diagnosis of breast abnormality," Machine Vision and Applications, vol. 30, 10 2019.

[15] A. F. Logullo, K. C. Prigenzi, C. C. Nimir, A. F. Franco, and M. S. Campos, "Breast microcalcifications: Past, present and future," Molecular and clinical oncology, vol. 16, no. 4, p. 81, 2022.

[16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 618–626, 2017.

[17] G. Cardenosa, Breast Imaging. Core curriculum, Lippincott Williams & Wilkins, 2004.